

Assignment 6: Mining and Proof-of-Work Lab

BLCH9X2 Blockchain |

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1. Purpose, and how to read this report alongside the notebook

This report accompanies the notebook **Assignment6_Bukho_Ndziweni_Blockchain_220081696**, submitted as HyperText Markup Language (HTML) with all outputs preserved, together with **mining.py**. Every claim below names the cell that produced it, using the execution numbers In [1] to In [6], and quotes the value that cell actually printed. The code was produced with the assistance of Artificial Intelligence and is commented thoroughly for clear understanding and interpretation of results.

Every timing figure in this report was measured on my own machine. Attempt counts, nonces and hashes are deterministic and reproduce anywhere, because timestamps are fixed constants and every nonce search starts at zero. Wall-clock seconds and hash rates are not: they were measured on the machine that ran the notebook, and are reported exactly as printed.

Sections 2 to 5 answer Part A. Section 6 answers Part B. Section 7 answers Part C.

Cell	What it does	Brief item
In [1]	Target arithmetic: difficulty, target and the rule this lab defines	A (a)
In [2]	The miner, and one block mined and verified	A (a)
In [3]	Parameter study: five difficulty levels, five trials each	A (b)
In [4]	Block reward paid to a miner address, and an attempt to redirect it	A (c)
In [5]	Retargeting from a deliberately wrong difficulty	B
In [6]	Retargeting after the hash rate is cut, and why this lab fixes difficulty	B

2. Part A (a): the rule this lab defines

A SHA-256 (Secure Hash Algorithm, 256-bit) hash is 64 hexadecimal characters, which is a very large whole number between 0 and 2 to the power 256, less one. Mining searches for a nonce whose hash, read as that number, falls at or below a ceiling called the target. Difficulty is how many times smaller than the maximum that ceiling is, so the target equals the maximum divided by the difficulty.

The rule stated, as the brief requires: a block is mined when its hash, read as a number, is at or below MAX_TARGET divided by the difficulty. This is the numeric target form rather than a count of leading zeros.

Two reasons for that choice, and both earned their keep later. Part B multiplies the target by fractional amounts during retargeting, which a count of zeros cannot express, since zeros move only in steps of sixteen. And the expected number of attempts falls out exactly as the difficulty itself, because a hash is equally likely to land anywhere in the range, so the chance of landing at or below the maximum divided by d is one in d . That gives section 3 a theoretical column to measure against rather than an approximation.

In [1] confirms the two forms are the same idea. At difficulty 65,536 the target is 0000ffff..., so a hash must begin with four zeros, which is exactly the leading-zeros rule used in Assignment 5. The difference shows at difficulty 16,384, where the target is 0003ffff.... Section 4 mined a block whose

winning hash begins 0003cbbd, a value that is valid under the numeric rule and inexpressible under a leading-zeros one.

3. Part A (b): parameter study of attempts and wall-clock time

There were five difficulty levels tested, with each level being four times more difficult than the previous one. Five distinct blocks were mined for each of the difficulty levels. Each mining operation involves the mining of different headers, as changing the same data set would yield only one nonce.

Difficulty	Expected attempts	Mean attempts	Median attempts	Mean seconds	Hashes per second
1,024	1,024	1,155	671	0.017	67,498
4,096	4,096	4,594	3,290	0.069	66,747
16,384	16,384	13,652	13,639	0.206	66,355
65,536	65,536	51,113	18,359	0.787	64,923
262,144	262,144	281,999	208,667	4.200	67,143

The whole study computed 1,762,564 hashes in 26.39 seconds, an overall rate of 66,777 hashes per second.

Mean time per block tracks difficulty closely. An approximately fourfold increase in difficulty leads to an approximately fourfold increase in mean seconds, starting at 0.017 seconds, 0.069, 0.206, 0.787, and ending at 4.200 seconds. That's how it's supposed to work because the hash rate is a characteristic of the CPU, while difficulty is irrelevant to it.

Attempt counts are far noisier than the means suggest. The five individual trials at difficulty 65,536 needed 1,313, 2,438, 39,395, 18,359 and 194,061 attempts, a spread of roughly 148 times within one difficulty level. That is why the median of 18,359 sits so far below the mean of 51,113. Mining attempts follow a geometric distribution, whose standard deviation is approximately equal to its mean, so five samples cannot pin the average down and the gap between the mean and expected columns should be read as sampling noise rather than as a defect in the rule.

The hash rate column is the internal consistency check. It covers the range 64,923 to 67,143 per second, which is 4 percent. This shows that the only thing that changes between levels is difficulty. A previous run of this experiment showed a completely different picture, starting at 46,569 and ending up at 68,262 per second, which is a range of 47 percent. This was due to the fact that this run started when the processor was cold; it took 0.12 seconds for the easy level to be finished, and the processor hasn't yet warmed up from the idle clock speed, whereas the hardest level takes more than 20 seconds under full load. The single block in In [2] corroborates this statement, showing 50,959 hashes per second from a cold start, against 66,868 for the same difficulty.

4. Part A (c): the block reward

The reward is paid by a coinbase transaction placed first in the block. It has no sender, because the units it pays did not exist before the block was found, and it pays a fixed subsidy of 50.00 units plus the fees collected from the payments in that block, set at 0.25 units each. Three blocks were mined at difficulty 16,384 and credited to the address MINER-BK-01.

Block	Nonce	Attempts	Seconds	Fees	Reward	Hash (first 16)
1	6	7	0.000	0.50	50.50	0001a42b68b4232d
2	20,554	20,555	0.449	0.25	50.25	0003cbbdd53fc063

Block	Nonce	Attempts	Seconds	Fees	Reward	Hash (first 16)
3	13,316	13,317	0.333	0.50	50.50	00003a34582836f4

The miner earned 151.25 units in total, of which 150.00 is newly created subsidy and 1.25 is fees. Sending addresses go negative in the resulting ledger, because no balance check is performed: this lab measures mining rather than transaction validity.

Block 1 was found in seven attempts. Against an expectation of 16,384 that looks wrong, and it is not. Expected attempts is an average over many blocks, not a quota, and any individual block can be found immediately. It is the same geometric distribution that produced the spread in section 3, seen from its lucky end.

The reward is protected by the proof of work rather than by a rule forbidding edits. Redirecting block 1's coinbase to another address after mining changed its hash from 0001a42b68b4232d... to bd743d194e737634..., which no longer meets the target. A thief would have to redo the search, and having done so would be mining their own block rather than stealing this one.

5. Part A (d): proof of work against proof of stake for a regulated financial market infrastructure

A financial market infrastructure (FMI) is a payment system, central securities depository, securities settlement system, central counterparty or trade repository, supervised against the Principles for Financial Market Infrastructures issued by the Committee on Payments and Market Infrastructures (CPMI) and the International Organization of Securities Commissions (IOSCO). The question of which consensus mechanism suits such an institution is usually argued on energy. That is the least interesting of the three dimensions below.

5.1 Security

Proof of work (PoW) purchases security externally. The adversary must purchase and operate hardware, and the threat is the cost, made upfront and paid over as long as the adversary needs to maintain a competitive advantage. The lab gives physical form to this math: at a difficulty of 262,144, the cost per block was an average of 281,999 hashes and 4.200 seconds on one thread. A network is the same math on multiple threads, and the budget for security is measurable externally as the total hash rate.

PoS ties security to the network in terms of stake. The validator invests some money, and in case of any bad behavior, he is penalized in terms of slashing, meaning that the deterrent is loss after the activity rather than an expenditure prior to that. The work done by Saleh (2021) is significant as she offers the first economic model of PoS consensus process and proves that an appropriately small reward structure not only makes possible consensus but also excludes any forking equilibrium. It is important as it solves nothing-at-stake problem: the validator himself is a stakeholder who knows the consequences of his actions on the asset's value.

For a supervised institution the difference that bites is not which is stronger but which is measurable and which is stable. A PoW security budget is external and observable. A PoS security budget is denominated in the staked asset, so the cost of attacking the system falls exactly when that asset falls, which is to say in precisely the conditions where an FMI is most likely to be tested. That is a procyclical security budget, and a supervisor accustomed to countercyclical capital requirements should find it uncomfortable.

The other distinguishing factor is in favor of PoS. With supervision comes accountability, which in turn requires that one identifies those who are responsible for the operations being supervised. The anonymous miner will not be licensed, monitored, or sanctioned, while stakeholders can easily be permissioned and linked to their identity without modifying any part of the consensus process.

5.2 Energy

The energy consumed in PoW systems is not a mere by-product that can be optimized out of existence. This energy consumption is the very mechanism through which the proof-of-work is executed, and without it there will be nothing for the attack to compete against. Thus, the lab has spent 1,762,564 hashes and 26.39 seconds on mining worthless blocks, and their energy expenditure has been the goal of their operation, not its side effect.

Now the scale of this alternative can actually be quantified, not discussed. The transition of Ethereum to PoS that occurred on 15 September 2022 has cut the annual energy consumption of the network by approximately 99.95 percent according to the Ethereum Foundation, and more than 99.988 percent according to Crypto Carbon Ratings Institute, from 23 million megawatt hours per year to more than 2,600. Whatever the exact numbers, the difference is an order of magnitude.

The argument that matters for an FMI, though, is not environmental. It is that PoW's energy bill purchases sybil resistance in an open market, and an FMI has no use for that property. Its participants are licensed institutions, known in advance and admitted by contract and regulation. Paying for permissionless entry, and paying for it continuously, is paying for something the institution is in fact obliged not to want.

5.3 Settlement finality

In line with Principle 8 of the PFMI, an FMI is supposed to ensure that final settlement is achieved in an unambiguous and unqualified manner, at least by the end of the value date, or intraday where necessary. In this context, final settlement implies an irrevocable and unconditional transfer or discharge of an obligation.

PoW can't deliver this. Nakamoto (2008) is clear on this point – that the probability of success is diminishing with the number of blocks and the formula for that decline is found in section 11 of the whitepaper but it never hits zero. In assignment 5, the reverse was priced, but still measuring the same thing – the cost of undoing two blocks rather than ruling it out. The declaration that undoing becomes increasingly expensive is different in kind to the declaration that it isn't possible anymore. There is no number of confirmations that will bridge this gap.

PoS implementations that have a clear “finality” gadget would fit the bill better. Finality is achieved by Ethereum approximately every two epochs, which means that once in about 12.8 minutes. Reversing finality can be done only if at least one third of all the staked money is burnt. This is, of course, not absolute finality, but it gives both the moment and the price for reversing, and thus it is much easier to codify into the rules..

None of the two processes in itself guarantees legal finality. Legal finality lies in the law's designation, and the protocol may give at most an opportunity for the law to point out something. The question here is whether the design can provide an instant that is traceable to certain parties which the legal process can make irrevocable. Proof-of-stake with the finality gadget does; open proof-of-work doesn't.

5.4 Conclusion

For a regulated FMI, PoS is the better of the two on all three dimensions, but the honest conclusion is that the comparison is the wrong one. Both mechanisms exist to solve consensus among parties whose identities are unknown, and an FMI knows exactly who its participants are. Once the validator set is definable, a permissioned Byzantine fault tolerant design of the kind set out by Castro and Liskov (1999) delivers deterministic finality in a single agreement round, at a latency suited to real time gross settlement (RTGS), without buying open participation at all. The relevant question for an institution such as the South African Reserve Bank's settlement system is therefore not PoW against PoS, but what a distributed ledger of any kind adds over a well-run RTGS, and the answer has to be found in participant experience and resilience rather than in the consensus mechanism.

6. Part B: difficulty retargeting, and why this lab fixes difficulty

Bitcoin measures how long the last 2,016 blocks took and multiplies the target by the ratio of actual to expected time, clamped to a factor of four in either direction. In [5] applies the same arithmetic at laboratory scale, with epochs of 5 blocks and a target block time of 0.20 seconds, starting from a difficulty set far too low on purpose.

Epoch	Difficulty	Seconds per block	Raw ratio	Next difficulty
1	256	0.0034	58.89	1,024
2	1,024	0.0128	15.62	4,096
3	4,096	0.0562	3.56	14,572
4	14,572	0.0686	2.92	42,489
5	42,489	0.8180	0.24	10,622
6	10,622	0.0877	2.28	24,234
7	24,234	0.3810	0.52	12,721
8	12,721	0.2327	0.86	10,933

The mean hash rate across the run was 56,562 per second, so the difficulty that fits this machine is 11,312. The algorithm finished at 10,933, within 3 percent of it.

Epochs 1 and 2 show the clamp binding upwards. A 58 times increase was required by the raw ratio, but the algorithm would not allow that to happen, opting instead for a multiplication by 4. Epoch 5 also illustrates the lower bound when decreasing difficulty; since the block time was 0.8180 seconds, the resulting ratio was 0.24, less than the quarter bound, hence making the difficulty go down to 10,622 instead of 10,203. The clamp is the thing that prevents one weird epoch or a miner's fake timestamp from moving difficulty wildly in one step.

After convergence the difficulty oscillates rather than settling. It moves between 10,622 and 24,234 around a correct value of 11,312. This is the small sample problem from section 3 arriving in a new place: five blocks is far too few to measure a block time, so the algorithm is partly chasing noise. Bitcoin averages over 2,016 blocks, roughly two weeks of evidence, for precisely this reason.

6.1 Retargeting when the hash rate changes

In [6], the reduction of the effective hash rate occurs mid-way through the computation process. This is not just for show because each effort incurs additional overhead of doing an extra hash that is as expensive as any other. Epochs 1–3 are done as fast as possible while Epochs 4–8 are done slowly.

Epoch	Speed	Difficulty	Seconds per block	Hashes per second	Next difficulty
1	full	11,312	0.1478	52,913	15,310
2	full	15,310	0.2899	55,316	10,562
3	full	10,562	0.1465	55,995	14,419
4	halved	14,419	0.2047	28,842	14,091
5	halved	14,091	0.3781	28,086	7,452
6	halved	7,452	0.3863	29,447	3,858
7	halved	3,858	0.1211	29,349	6,374
8	halved	6,374	0.2002	28,821	6,368

The measured rate fell from 54,741 to 28,909 hashes per second, a reduction of 47 percent rather than the intended 50, which is the difference between a measurement and an assumption. Mean block time was 0.1947 seconds before the change and 0.2581 after, against a target of 0.2000, so the algorithm held its aim while the ground moved. The difficulty settled near 6,368, roughly half its starting value, which is the correct response to half the power.

The conclusion is worth stating carefully. Losing half the mining power does not halve the speed of the chain and does not halve its usefulness. It costs one slow epoch, after which block times recover at a lower difficulty. What changes permanently is the security budget, because half as much work now stands behind each block, which is the same quantity the essay in section 5 identifies as the thing a supervisor would have to monitor.

6.2 Why this lab uses a fixed difficulty

Reproducibility comes first. With a fixed difficulty and fixed timestamps, every nonce and hash in the notebook is identical on any machine. Retargeting makes the difficulty a function of local processor speed, so nobody else could reproduce the values, only the shape of the behaviour, as the difference between my two runs of section 6 already shows.

The parameter study needs one variable at a time. Section 3 measures how attempts and time respond to difficulty, and if the difficulty moved during the run nothing could be attributed to anything.

The network itself and the adversarial clock also do not apply to the single process with fixed timestamps. The reason for which is that Bitcoin retargets on the timestamps provided by miners who can cheat. That's what makes clamp and median-of-eleven-timestamp rules necessary. Therefore, section 6 describes the math of retargeting and does not describe the adversarial problem it solves.

Finally, epochs consisting of five blocks are too small to measure anything as the oscillations above prove. It is necessary to fix the difficulty so that it would not report noise as the result.

7. Part C: open questions for a regulated FMI

The essay in section 5 concludes that PoS suits a regulated FMI better than PoW on security, energy and finality, and that a permissioned design suits it better than either. Each of those conclusions rests on assumptions that a supervisor would have to test rather than accept. The following are open, in the sense that the literature cited here does not settle them.

1. At what moment does protocol finality become legal finality? Castro and Liskov (1999) deliver a commit that is deterministic under their fault bound, and Ethereum's finality gadget names a

checkpoint, but legal irrevocability is conferred by designation in law rather than by a protocol. Can a probabilistic rule of the kind Nakamoto (2008) describes ever be designated, and if not, is PoW categorically excluded from FMI use rather than merely ill-suited?

2. How should a supervisor stress-test a security budget in terms of the asset in which it is denominated? If the cost of attacking a PoS mechanism declines with the price of the token being used to stake, the mechanism is procyclical. What is the equivalent of a capital requirement in PoS? What should a stress test for such a requirement look like?
3. Does automated slashing fit with administrative due process? The confiscation of capital from a licensed entity through automated means, without a hearing, is somewhat at odds with the due process considerations that usually apply to such entities. By whom is an erroneous slash determined if the error is an operational one?
4. What measure should control validator concentration? According to Saleh (2021), validators are modelled as stakeholders; however, with custodial staking, a small number of service providers gain control, thus creating the same concentration problem that decentralization was designed to solve. At what point should supervision become necessary, and relative to what denominator?
5. How should energy be disclosed? Per transaction is the common metric and arguably the least meaningful, since it varies with block occupancy rather than with anything about the mechanism. Energy per unit of value settled, or per unit of security purchased, would be more informative and neither is standard.
6. How does an FMI running on a public chain execute an orderly recovery or wind-down of infrastructure it does not control? The PFMI expect a credible plan, and a plan that depends on the cooperation of an anonymous validator set is difficult to call credible.
7. Finally, and most awkwardly for the whole exercise: if the conclusion is a permissioned ledger with known validators, what measurable advantage does it retain over a modern RTGS? The consensus literature answers how to agree among strangers, and an FMI has no strangers.

8. References

Antonopoulos, A. M. and Harding, D. A. (2023) *Mastering Bitcoin: Programming the Open Blockchain*. 3rd edn. Sebastopol: O'Reilly Media. Mining and difficulty retargeting are covered in the mining chapter.

Castro, M. and Liskov, B. (1999) 'Practical Byzantine fault tolerance', *Proceedings of the Third Symposium on Operating Systems Design and Implementation*. New Orleans, February 1999.

CPSS-IOSCO (2012) *Principles for Financial Market Infrastructures*. Basel: Bank for International Settlements. Principle 8, settlement finality.

Ethereum Foundation (2022) *The Merge*. Available at ethereum.org. Energy consumption reduction of approximately 99.95 percent, with the Crypto Carbon Ratings Institute measurement of over 99.988 percent reported separately.

Nakamoto, S. (2008) *Bitcoin: A Peer-to-Peer Electronic Cash System*. Section 11, calculations on the probability of an attacker catching up.

Saleh, F. (2021) 'Blockchain without waste: proof-of-stake', *The Review of Financial Studies*, 34(3), pp. 1156-1190.

van Flymen, D. (2017) 'Learn Blockchains by Building One'. Chain construction and verification.

9. Code repository

<https://github.com/bukhondziweni19/blockchain-assignment-6>